# Raphaël Honigsberg

PhD candidate ; Microfluidics, microfabrication & microbial imaging - ENS Paris-Saclay
Paris, France ; raphaelhonigsberg1@gmail.com ; linkedin.com/in/raphaël-honigsberg-307b9a310

**Seeking a postdoctoral position** from January-February 2027 - microfluidics, 3D microfabrication, single-cell imaging. Academic, open to industry.

## PROFILE

Biologist trained in microfluidics. I design and build microfluidic platforms end to end (CAD, 3D-printed molds, clean-room processes, live imaging and automated image analysis) to follow microbial growth down to single cells.

Next goal: a **scalable fabrication strategy for functional 3D microfluidics**, in an academic lab or with industry.

## RESEARCH EXPERIENCE

**2023 - Jan. 2027** — **PhD candidate - LBPA & LuMIn, ENS Paris-Saclay, CNRS, Université Paris-Saclay**

*Supervisors: E. Deprez, B. Le Pioufle, S. Bensalem ; Competitive doctoral contract (CDSN)*

- *Thesis:* development of a microfluidic system to study microalgae-bacteria interactions, to optimise biofuel production yields
- Designed a pump-free, pipette-operated PDMS chip with **400 gravity-trap wells**, cast from a single 3D-printed mold (no clean room)
- Built on-chip **light × chemical gradient screening**: 100 conditions × 4 replicates per chip, up to 9 chips per run
- Multi-day time-lapse epifluorescence and confocal imaging of *Chlorella vulgaris* and co-cultured bacteria, down to single cells; automated acquisition in NIS-Elements
- Image-analysis pipeline (StarDist segmentation, growth-rate fitting) on millions of segmented cells; COMSOL modelling of flow, shear and diffusion

**Jan. - Jun. 2023** — **Master's thesis - Institut Pasteur (G.-F. Richard) & LadHyX, École Polytechnique (C. N. Baroud)**

- Single-cell analysis of the checkpoint response to DNA double-strand breaks in *S. cerevisiae*
- Designed a dedicated microfluidic chip; fluorescence microscopy and image analysis

**Oct. 2021 - Jul. 2022** — **Pre-doctoral research year (ARPE) - Faherty Lab, Harvard Medical School & Massachusetts General Hospital, Boston**

- Adherence factors of *Shigella flexneri* (BSL-2): infection assays on HT-29 monolayers, λ-Red mutagenesis, cloning, biofilm assays - Co-author, *Gut Microbes* 2024
- Supervised two students (one graduate, one master's)

**2019 - 2020** — **Research internships - Institut de Chimie Physique, UMR 8000, Université Paris-Saclay (M. Erard)**

- Fluorescent proteins for labelling cellular endomembranes: cloning, transfection, FRET-FLIM microscopy

## INNOVATION & AWARDS

**2025** — **Créativité Prize**, Starthèse start'up pitch-ID, Université Paris-Saclay - project *Honigs*, 3D-printed microfluidics for cell culture ; selected to pitch at **BIG 2025** (Bpifrance), Paris

**2023** — Competitive doctoral contract for ENS graduates (CDSN)

## TEACHING & SUPERVISION

**2021 - 2022** — Supervision of two students (graduate and master's), Harvard Medical School / MGH

**2 years** — Oral examiner (*khôlles*) in biology, CPGE BCPST, Paris - 1 h/week ; exam grading

## PUBLICATIONS & COMMUNICATIONS

León Y, **Honigsberg R**, Rasko DA, Faherty CS. Gastrointestinal signals in supplemented media reveal a role in adherence for the *Shigella flexneri* sap autotransporter gene. *Gut Microbes* 16(1), 2024. doi:10.1080/19490976.2024.2331985

**Honigsberg R**, et al. 3D-printed mould for a through-hole array: a pump-free chip for light × nitrate phenotyping of microalgae (working title). In preparation for *Lab on a Chip*.

**Honigsberg R**, Ghasemi R, Audibert JF, Deprez E, Le Pioufle B, Bensalem S. Low-cost microfluidic platform to monitor early growth from single microalgal cells in bacteria co-culture. **Poster, MicroTAS 2026**, Granada, Spain.

## EDUCATION

| | |
|---|---|
| 2023 - 2027 | **PhD**, Université Paris-Saclay, doctoral school SDSV (ED 577) - defence January 2027 |
| 2019 - 2023 | **École Normale Supérieure Paris-Saclay** - normalien (civil-servant student), ENS diploma 2023 |
| 2022 - 2023 | **MSc** Fundamental Microbiology, Université Paris Cité & Institut Pasteur |
| 2020 - 2021 | **MSc year 1** Biology & Health (Boris Ephrussi track), Université Paris-Saclay, with honours |
| 2019 - 2020 | **BSc** Biology & Health, Université Paris-Saclay, with honours |
| 2016 - 2019 | **Classes préparatoires** BCPST, Lycées Henri-IV and Fénelon, Paris - admitted to ENS Paris-Saclay (top 2 %) |

## SKILLS

| | |
|---|---|
| **Microfabrication** | CAD at expert level: Fusion 360 (molds, chips), L-Edit (mask layout) ; 3D printing (SLA, DLP) of molds and devices ; SU-8 photolithography, single- and multi-layer (Süss MJB4, RC8 spin coater, fully autonomous) ; Dektak profilometry ; PDMS soft lithography ; plasma bonding (Diener Atto/Femto) ; UV-ozone ; parylene deposition (SCS PDS 2010, full mechanical maintenance) |
| **Materials & surfaces** | Hydrogels for microfluidics: alginate blended with CMC, methylcellulose, xanthan or gum arabic (rheology), PEGDA ; gel-surface interactions and meniscus pinning to hold gel walls in channels ; silane chemistry and surface coatings (silanisation, parylene) |
| **Modelling** | COMSOL Multiphysics: laminar flow, shear stress, diffusion |
| **Imaging** | Epifluorescence and confocal microscopy ; NIS-Elements, automated multi-day acquisitions ; single-cell image analysis: Fiji, StarDist, Cellpose, ilastik, TrackMate |
| **Data & AI** | Python analysis pipelines (images, growth fits, statistics, PCA) built with AI coding assistants ; AI-assisted literature screening and task automation ; publication-quality figures |
| **Biology** | Bacteria (*E. coli*, *S. flexneri* BSL-2, *P. stutzeri*; wild marine bacteria isolated from oyster and periwinkle digestive tracts, screened for cellulolytic strains degrading microalgae), yeast, microalgae (*Chlorella*, *Nannochloropsis*; synchronised cultures), algae-bacteria co-cultures, human cell lines (HT-29) ; molecular biology: cloning, PCR, λ-Red mutagenesis |
| **Languages** | French (native) ; English (C1-C2, Cambridge Advanced; IELTS 8.0) ; Spanish (A2) |

## REFERENCES


**Dr Eric Deprez**
LBPA, CNRS, ENS Paris-Saclay
eric.deprez@ens-paris-saclay.fr

**Dr Sakina Bensalem**
ENS Paris-Saclay
sakina.bensalem@ens-paris-saclay.fr

**Prof. Christina S. Faherty**
Harvard Medical School & MGH, Boston
csfaherty@mgh.harvard.edu

**Prof. Bruno Le Pioufle**
LuMIn, ENS Paris-Saclay
bruno.le-pioufle@ens-paris-saclay.fr

**Prof. Charles N. Baroud**
LadHyX, École Polytechnique & Institut Pasteur
charles.baroud@pasteur.fr